Suggested Solutions

1	**(a)**	Loss in potential energy = mgh $= 1.32 \times 10^{12} \times 9.81 \times 5.0$ (since the average height of the water is 5.0 m) $= 6.5 \times 10^{13}$ J	
	(b)	Power from sea water = gravitational energy lost / time $= 6.5 \times 10^{13} / 6.0 \times 3600$ $= 3000 \times 10^{6}$ W Average power output $= 3000 \times 10^{6} \times 0.40$ = 1200 MW	
	(c)	There are valves within the dam that controls and regulates the flow of water into the turbines. {the higher the speed/ flow rate, the lower the time and therefore average power increases}	

2	**(a)**	The gravitational potential at a point is the work done per unit mass by an external agent in	